

2021Fall AMC12B**Problem 1**

What is the value of $1234 + 2341 + 3412 + 4123$?

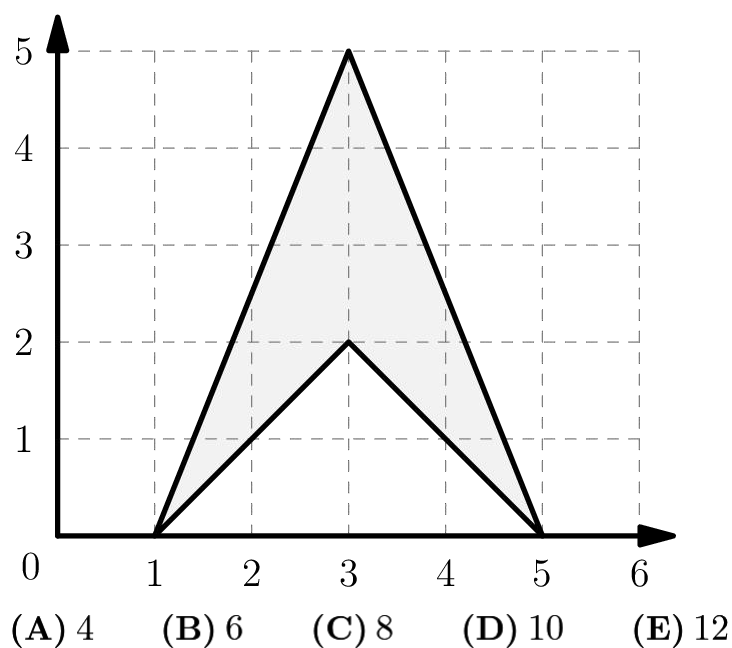
表达式 $1234 + 2341 + 3412 + 4123$ 的值是多少?

- (A) 10,000 (B) 10,010 (C) 10,110 (D) 11,000 (E) 11,110

Problem 2

What is the area of the shaded figure shown below?

下图中阴影部分的面积是多少?



Problem 3

At noon on a certain day, Minneapolis is N degrees warmer than St. Louis. At 4:00 the temperature in Minneapolis has fallen by 5 degrees while the temperature in St. Louis has risen by 3 degrees, at which time the temperatures in the two cities differ by 2 degrees. What is the product of all possible values of N ?

某一天的中午，Minneapolis 的气温比 St. Louis 高 N 度。4:00 时，Minneapolis 的气温下降了 5 度，而 St. Louis 的气温上升了 3 度，此时两个城市的气温相差 2 度。问 N 的所有可能值的乘积是多少？

- (A) 10 (B) 30 (C) 60 (D) 100 (E) 120

Problem 4

Let $n = 8^{2022}$. Which of the following is equal to $\frac{n}{4}$?

设 $n = 8^{2022}$ 。问以下哪一项等于 $\frac{n}{4}$ ？

- (A) 4^{1010} (B) 2^{2022} (C) 8^{2018} (D) 4^{3031} (E) 4^{3032}

Problem 5

Call a fraction $\frac{a}{b}$, not necessarily in the simplest form, special if a and b are positive integers whose sum is 15. How many distinct integers can be written as the sum of two, not necessarily different, special fractions?

如果 a 和 b 是正整数，总和为 15，那么分数 $\frac{a}{b}$ ，无论是否为最简形式，称为“特殊的”。问有多少个不同的整数可以写成两个特殊分数的和，这里的两个特殊分数不要求互异？

- (A) 9 (B) 10 (C) 11 (D) 12 (E) 13

Problem 6

The largest prime factor of 16384 is 2 because $16384 = 2^{14}$. What is the sum of the digits of the greatest prime number that is a divisor of 16383?

16,384 的约数中的最大素数是 2, 因为 $16,384 = 2^{14}$ 。问 16,383 的约数中的最大素数的各位数字之和是多少?

- (A) 3 (B) 7 (C) 10 (D) 16 (E) 22

Problem 7

Which of the following conditions is sufficient to guarantee that integers x , y , and z satisfy the equation $x(x - y) + y(y - z) + z(z - x) = 1$?

以下哪个条件可以确保整数 x 、 y 和 z 满足方程

$$x(x - y) + y(y - z) + z(z - x) = 1?$$

- (A) $x > y$ and $y = z$
(B) $x = y - 1$ and $y = z - 1$
(C) $x = z + 1$ and $y = x + 1$
(D) $x = z$ and $y - 1 = x$
(E) $x + y + z = 1$

Problem 8

The product of the lengths of the two congruent sides of an obtuse isosceles triangle is equal to the product of the base and twice the triangle's height to the base. What is the measure, in degrees, of the vertex angle of this triangle?

钝角等腰三角形两条相等的边长的乘积等于底边与三角形上底边高的两倍的乘积。问这个三角形的顶角是多少度?

- (A) 105 (B) 120 (C) 135 (D) 150 (E) 165

Problem 9

Triangle ABC is equilateral with side length 6. Suppose that O is the center of the inscribed circle of this triangle. What is the area of the circle passing through A , O , and C ?

三角形 ABC 是边长为 6 的等边三角形。假设 O 是这个三角形的内切圆圆心。问经过 A 、 O 和 C 的圆的面积是多少?

- (A) 9π (B) 12π (C) 18π (D) 24π (E) 27π

Problem 10

What is the sum of all possible values of t between 0 and 360 such that the triangle in the coordinate plane whose vertices are $(\cos 40^\circ, \sin 40^\circ)$, $(\cos 60^\circ, \sin 60^\circ)$, and $(\cos t^\circ, \sin t^\circ)$ is isosceles?

介于 0 和 360 之间, 并且使得坐标平面中顶点为 $(\cos 40^\circ, \sin 40^\circ)$, $(\cos 60^\circ, \sin 60^\circ)$ 和 $(\cos t^\circ, \sin t^\circ)$ 的三角形是等腰的所有可能 t 值的总和是多少?

- (A) 100 (B) 150 (C) 330 (D) 360 (E) 380

Problem 11

Una rolls 6 standard 6-sided dice simultaneously and calculates the product of the 6 numbers obtained. What is the probability that the product is divisible by 4?

Una 同时掷出 6 个标准的有 6 个面的骰子，并计算得到的 6 个数的乘积。问乘积能被 4 整除的概率是多少？

- (A) $\frac{3}{4}$ (B) $\frac{57}{64}$ (C) $\frac{59}{64}$ (D) $\frac{187}{192}$ (E) $\frac{63}{64}$

Problem 12

For n a positive integer, let $f(n)$ be the quotient obtained when the sum of all positive divisors of n is divided by n . For example, $f(14) = (1 + 2 + 7 + 14) \div 14 = \frac{12}{7}$. What is $f(768) - f(384)$?

对于正整数 n ，令 $f(n)$ 为 n 的所有正约数之和除以 n 所得的商。例如， $f(14) = (1 + 2 + 7 + 14) \div 14 = \frac{12}{7}$ 。问 $f(768) - f(384)$ 是多少？

- (A) $\frac{1}{768}$ (B) $\frac{1}{192}$ (C) 1 (D) $\frac{4}{3}$ (E) $\frac{8}{3}$

Problem 13

Let $c = \frac{2\pi}{11}$. What is the value of $\frac{\sin 3c \cdot \sin 6c \cdot \sin 9c \cdot \sin 12c \cdot \sin 15c}{\sin c \cdot \sin 2c \cdot \sin 3c \cdot \sin 4c \cdot \sin 5c}$?

设 $c = \frac{2\pi}{11}$ 。表达式

$$\frac{\sin 3c \cdot \sin 6c \cdot \sin 9c \cdot \sin 12c \cdot \sin 15c}{\sin c \cdot \sin 2c \cdot \sin 3c \cdot \sin 4c \cdot \sin 5c}$$

的值是多少？

- (A) -1 (B) $-\frac{\sqrt{11}}{5}$ (C) $\frac{\sqrt{11}}{5}$ (D) $\frac{10}{11}$ (E) 1

Problem 14

Suppose that $P(z)$, $Q(z)$, and $R(z)$ are polynomials with real coefficients, having degrees 2, 3, and 6, respectively, and constant terms 1, 2, and 3, respectively. Let N be the number of distinct complex numbers z that satisfy the equation $P(z) \cdot Q(z) = R(z)$. What is the minimum possible value of N ?

假设 $P(z)$ 、 $Q(z)$ 和 $R(z)$ 是实系数多项式，次数分别为 2、3 和 6，常数项分别为 1、2 和 3。令 N 是满足方程 $P(z) \cdot Q(z) = R(z)$ 的不同复数 z 的个数。问 N 的最小可能值是多少？

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 5

Problem 15

Three identical square sheets of paper each with side length 6 are stacked on top of each other. The middle sheet is rotated clockwise 30° about its center and the top sheet is rotated clockwise 60° about its center, resulting in the 24-sided polygon shown in the figure below. The area of this polygon can be expressed in the form $a - b\sqrt{c}$, where a , b , and c are positive integers, and c is not divisible by the square of any prime. What is $a + b + c$?

三张相同的边长为 6 的正方形纸片叠放在一起。然后中间的纸片绕其中心顺时针旋转 30° ，最上面的纸片绕其中心顺时针旋转 60° ，形成如下图所示的 24 边形。这个多边形的面积可以用 $a - b\sqrt{c}$ 的形式表示，其中 a 、 b 和 c 是正整数，并且 c 不能被任何素数的平方整除。问 $a + b + c$ 是多少？

- (A) 75 (B) 93 (C) 96 (D) 129 (E) 147

Problem 16

Suppose a, b, c are positive integers such

that $a + b + c = 23$ and $\gcd(a, b) + \gcd(b, c) + \gcd(c, a) = 9$. What is the sum of all possible distinct values of $a^2 + b^2 + c^2$?

假设 a, b 和 c 是正整数, 满足条件 $a + b + c = 23$, 以及 $\gcd(a, b) + \gcd(b, c) + \gcd(c, a) = 9$ 。问 $a^2 + b^2 + c^2$ 的所有不同的可能值的总和是多少?

- (A) 259 (B) 438 (C) 516 (D) 625 (E) 687

Problem 17

A bug starts at a vertex of a grid made of equilateral triangles of side length 1. At each step the bug moves in one of the 6 possible directions along the grid lines randomly and independently with equal probability. What is the probability that after 5 moves the bug never will have been more than 1 unit away from the starting position?

一只虫子从由边长为 1 的等边三角形组成的网格的某个顶点出发。每一步, 虫子随机且独立的沿着网格线以 6 个可能的方向之一移动到相邻的顶点。问在 5 步移动后, 虫子距离起始位置从来没有超过 1 的概率是多少?

- (A) $\frac{13}{108}$ (B) $\frac{7}{54}$ (C) $\frac{29}{216}$ (D) $\frac{4}{27}$ (E) $\frac{1}{16}$

Problem 18

Set $u_0 = \frac{1}{4}$, and for $k \geq 0$ let u_{k+1} be determined by the recurrence $u_{k+1} = 2u_k - 2u_k^2$.

This sequence tends to a limit; call it L . What is the least value of k such that $|u_k - L| \leq \frac{1}{2^{1000}}$?

设 $u_0 = \frac{1}{4}$, 对于 $k \geq 0$, u_{k+1} 由递推式 $u_{k+1} = 2u_k - 2u_k^2$ 确定。这个数列趋于某个极限, 记为 L 。满足

$$|u_k - L| \leq \frac{1}{2^{1000}}$$

的 k 的最小值是多少?

- (A) 10 (B) 87 (C) 123 (D) 329 (E) 401

Problem 19

Regular polygons with 5, 6, 7, and 8 sides are inscribed in the same circle. No two of the polygons share a vertex, and no three of their sides intersect at a common point. At how many points inside the circle do two of their sides intersect?

有 5、6、7 和 8 条边的正多边形都内接在同一个圆中。任意两个多边形没有公共顶点, 并且也没有三条多边形的边相交于一点。由这些多边形的两条边在圆内形成的交点有多少个?

- (A) 52 (B) 56 (C) 60 (D) 64 (E) 68

Problem 20

A cube is constructed from 4 white unit cubes and 4 blue unit cubes. How many different ways are there to construct the $2 \times 2 \times 2$ cube using these smaller cubes? (Two constructions are considered the same if one can be rotated to match the other.)

一个立方体由 4 个白色单位立方体和 4 个蓝色单位立方体构成。使用这些较小的立方体来构造该 $2 \times 2 \times 2$ 的立方体, 共有多少种不同的方法? (如果一种构型可以通过旋转成为另一种构型, 则认为两种构型是相同的。)

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Problem 21

For real numbers x ,

$$\text{let } P(x) = 1 + \cos(x) + i \sin(x) - \cos(2x) - i \sin(2x) + \cos(3x) + i \sin(3x) \text{ where } i = \sqrt{-1}.$$

For how many values of x with $0 \leq x < 2\pi$ does $P(x) = 0$?

对于实数 x ，设

$$P(x) = 1 + \cos(x) + i \sin(x) - \cos(2x) - i \sin(2x) + \cos(3x) + i \sin(3x),$$

其中 $i = \sqrt{-1}$ 。在 $0 \leq x < 2\pi$ 中，有多少个 x 值满足 $P(x) = 0$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Problem 22

Right triangle ABC has side lengths $BC = 6$, $AC = 8$, and $AB = 10$.

A circle centered at O is tangent to line BC at B and passes through A . A circle centered at P is tangent to line AC at A and passes through B . What is OP ?

直角三角形 ABC 的边长为 $BC = 6$, $AC = 8$, $AB = 10$ 。以 O 为圆心的圆与直线 BC 相切于 B ，并且经过点 A 。以 P 为圆心的圆与直线 AC 相切于 A ，并且经过点 B 。问 OP 的长度是多少?

- (A) $\frac{23}{8}$ (B) $\frac{29}{10}$ (C) $\frac{35}{12}$ (D) $\frac{73}{25}$ (E) 3

Problem 23

What is the average number of pairs of consecutive integers in a randomly selected subset of 5 distinct integers chosen from the set $\{1, 2, 3, \dots, 30\}$? (For example the set $\{1, 17, 18, 19, 30\}$ has 2 pairs of consecutive integers.)

考虑从集合 $\{1, 2, 3, \dots, 30\}$ 中随机选择 5 个不同整数组成的子集，其中连续整数对数目的平均值是多少？（例如，集合 $\{1, 17, 18, 19, 30\}$ 中有 2 对连续整数。）

- (A) $\frac{2}{3}$ (B) $\frac{29}{36}$ (C) $\frac{5}{6}$ (D) $\frac{29}{30}$ (E) 1

Problem 24

Triangle ABC has side lengths $AB = 11$, $BC = 24$, and $CA = 20$. The bisector of $\angle BAC$ intersects $\overline{BC}$ in point D , and intersects the circumcircle of $\triangle ABC$ in point $E \neq A$. The circumcircle of $\triangle BED$ intersects the line AB in points B and $F \neq B$. What is CF ?

三角形 ABC 的边长为 $AB = 11$, $BC = 24$, $CA = 20$ 。 $\angle BAC$ 的平分线与 $\overline{BC}$ 相交于点 D ，并且与 $\triangle ABC$ 的外接圆相交于点 $E \neq A$ 。 $\triangle BED$ 的外接圆与线 AB 相交于点 B 和 $F \neq B$ 。问 CF 的长度是多少？

- (A) 28 (B) $20\sqrt{2}$ (C) 30 (D) 32 (E) $20\sqrt{3}$

Problem 25

For n a positive integer, let $R(n)$ be the sum of the remainders when n is divided by 2, 3, 4, 5, 6, 7, 8, 9, and 10. For example, $R(15) = 1 + 0 + 3 + 0 + 3 + 1 + 7 + 6 + 5 = 26$. How many two-digit positive integers n satisfy $R(n) = R(n + 1)$?

对于正整数 n ，令 $R(n)$ 为 n 除以 2、3、4、5、6、7、8、9 和 10 的余数之和。例如， $R(15) = 1 + 0 + 3 + 0 + 3 + 1 + 7 + 6 + 5 = 26$ 。问有多少个两位正整数 n 满足 $R(n) = R(n + 1)$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

2021Fall AMC12B Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13
E	B	C	E	C	C	D	D	B	E	C	B	E
14	15	16	17	18	19	20	21	22	23	24	25	
B	E	B	A	A	E	A	A	C	A	C	C	